

Cambodia-focused TikTok Shop Lite MVP

Executive summary

A Cambodia-focused “TikTok Shop lite” MVP is best treated as a **social-commerce marketplace** rather than a pure content platform: discovery feed → product page → checkout (COD + KHQR) → fulfilment/tracking → disputes/returns → seller payout. The “lite” constraint should primarily remove (at launch) heavy features such as creator marketplaces, multi-warehouse fulfilment, and complex advertising systems, while still delivering **trust primitives** (seller verification, product policy enforcement, order-state integrity, and a payout hold/settlement mechanism). TikTok Shop itself operationalises these primitives with explicit fulfilment SLAs and enforcement (including cancellation risk for late fulfilment) and with settlement schedules/tiers and risk-based holds/reserves—these concepts map well to a first-principles Cambodia build, even if the exact timings and tools differ. ¹

Several material parameters are **unspecified**: budget, team size, exact launch date, target categories (e.g., beauty vs electronics vs FMCG), whether you will host user-generated video at launch, and the domain/TLD strategy. Where these affect timelines and scope, this report provides **conditional, stack-agnostic recommendations** and defines acceptance criteria and success metrics that can be tuned once constraints are fixed.

Regulatory and compliance design can't be bolted on later. Cambodia's **Law on Electronic Commerce** requires intermediaries/e-commerce service providers to obtain an e-commerce permit/licence (and points to an online service certificate role for the telecom ministry), mandates consumer-facing disclosures for online selling (including payment/cancellation/return/refund terms), and contains personal-data security duties. ² It also states that operating/providing electronic payment services to the public requires prior authorisation or licence from the central bank, creating a strong incentive to architect the MVP so that **licensed payment providers** handle regulated payment functions while your platform focuses on **commerce orchestration and ledgers**. ³

Cambodia-tailored product definition and brand options

Product scope for a “TikTok Shop lite” MVP in Cambodia

The platform should ship as three tightly-scoped surfaces that share one order/ledger core:

Buyer surface (mobile-first web or app)

A vertical discovery feed can start as product cards (images + short clips) with optional “creator-style” content later. The core is: browse → product detail → checkout → order tracking. The experience should be bilingual (Khmer/English) if you expect cross-border or tourist buyers; Khmer-first is often necessary for mass adoption, and Cambodia's consumer information standards can require Khmer-language inclusion in written consumer information standards under regulator prakas. ⁴

Seller surface (web dashboard, optional mobile lite)

The seller must be able to onboard, submit documents, publish products, manage inventory, accept/cancel orders, generate shipping actions (pickup/drop-off), and see payout status. TikTok Shop's seller

guidance emphasises that fulfilment is operationally central (order confirmation → packing → dispatch/tracking updates), and that SLA failures can trigger enforcement. ⁵

Admin/ops surface

You need a moderation and risk-control console from day one because Cambodian law assigns responsibilities to intermediaries once they become aware of illegal activity/content and outlines actions like removal and notification/preservation in certain circumstances. ⁶

A practical “lite” cut removes: a full creator affiliate marketplace, paid ads, warehouse fulfilment, complex recommendation ML, and cross-border logistics. It retains: strong order-state machine, payouts/holds, and basic content moderation.

Brand/name shortlist comparison

The domain availability check is **unspecified and not performed** (no TLD preference provided; domain availability is time-variant). The “legal risk” column is not legal advice; it is an operational risk assessment (confusability, descriptive/generic risk, likely trademark friction, and closeness to TikTok/TikTok Shop naming patterns).

Name	Brand risk (look/feel similarity to TikTok)	Memorability	Pronunciation (Khmer/English)	Legal risk (operational assessment)	Domain availability
KhmerCart	Low	Medium-High	Easy	Medium (descriptive/ common words)	Unspecified / not checked
ShopSrok	Low	High	Medium (non-Khmer speakers may stumble)	Low-Medium (distinctive; uses common Khmer term)	Unspecified / not checked
KlikKhmer	Low	High	Easy-Medium	Medium (common “click” motif; fairly distinctive)	Unspecified / not checked
PsarNow	Low	High	Medium (requires familiarity with “psar”)	Medium (uses common Khmer market term; still distinctive)	Unspecified / not checked
TrendCart	Low	Medium	Easy	Medium (generic; “Trend” widely used)	Unspecified / not checked
LiveCart	Low	Medium	Easy	Medium (generic; “Live” widely used)	Unspecified / not checked

Name	Brand risk (look/feel similarity to TikTok)	Memorability	Pronunciation (Khmer/English)	Legal risk (operational assessment)	Domain availability
ClipCart	Low	Medium	Easy	Medium (generic; may conflict with prior “Clip” marks)	Unspecified / not checked
BuySrok	Low	Medium	Medium	Low-Medium	Unspecified / not checked
SrokMall	Low	Medium	Medium	Medium (generic marketplace connotation)	Unspecified / not checked
PopCart	Low	Medium	Easy	Medium (generic; “Pop” is crowded)	Unspecified / not checked

Interpretation criteria

“Srok” (ស្រុក) commonly conveys homeland/local area/country and can signal “local commerce” rather than a foreign clone; it may also skew slightly traditional depending on brand design. The safest direction is a name that does not include “Tok/Tik” and does not attempt to mimic TikTok’s mark.

Legal and compliance requirements tailored to Cambodia

E-commerce licensing and intermediary obligations

Cambodia’s **Law on Electronic Commerce** explicitly states that an intermediary/e-commerce service provider must request a permission letter or licence from the Ministry of Commerce ⁷ and the Ministry of Posts and Telecommunications ⁸, distinguishes a permission letter for natural persons and a licence for legal persons, and notes that the telecom ministry shall provide an online service certificate, with further exceptions/types/procedures to be determined by sub-decree. ⁹

The Ministry of Commerce has issued public notices (English translations exist) stating that individuals, sole proprietors, legal entities, and branches of foreign companies conducting business via electronic platforms are required to apply for an e-commerce permit/licence via the department of business registration or the e-commerce licensing portal, and that operating without a permit/licence can lead to cessation of operations and monetary fines. ¹⁰

For a marketplace MVP, treat yourself as an “intermediary” in practical terms: design operational processes for notice-and-action, preservation/audit trails, and escalation. The law provides a framework where intermediaries that become aware of unlawfulness/illegality must take steps (remove content, preserve evidence, notify competent ministries) and may have their liability shaped by such actions. ⁶

Consumer protection and product information duties

The law's consumer protection chapter requires that anyone selling goods or services to consumers via electronic communications provide accurate, clear information at minimum including: seller identity and address/contact, effective communication channel, the terms/conditions/costs including payment method, and details on cancellation/withdrawal, return/exchange and refunds, plus a description of the actual goods/services. ¹¹

Cambodia's consumer protection framework (as published via the Digital Economy and Business Committee's legal framework downloads and widely circulated official translations) prohibits misleading representations in supplying/promoting goods/services and establishes information-standard obligations for businesses supplying or advertising goods/services, with further information standards to be set by regulator prakas. ¹²

Practical MVP implication: your listing schema and UI must support **mandatory consumer disclosures**, and your moderation tooling must be able to enforce them (e.g., reject listings missing price/currency, return policy, seller address/contact channel).

Data protection and security duties

The Law on Electronic Commerce includes an explicit "Data protection" article requiring persons holding personal information in electronic form to use reasonable security safeguards to avoid loss, unauthorised access/use/modification/leak/disclosure, except with permission or as authorised by law; it also prohibits bad-faith unauthorised interference/access/copy/extract/delete/modify. ¹³

MVP implication: at minimum, implement encryption in transit (TLS), strict access controls, audit logging for admin actions, and secure storage/retention policies aligned to legal retention obligations for records (the law also addresses retention of electronic records in evidentiary contexts). ¹⁴

Payments licensing risk and safe MVP architecture

The Law on Electronic Commerce states that "any person operating a payment system, providing a payment service, issuing an electronic payment instrument or providing an electronic means of payment" considered electronic payment must obtain prior authorisation or licence from the National Bank of Cambodia ¹⁵, and that the central bank is the competent authority to formulate regulations relating to electronic payment. ¹⁶

This is the core compliance design choice:

- **Lower-regulatory surface MVP:** integrate licensed payment providers as merchant-of-record/payment processors; your platform records orders and maintains an internal ledger, but does not itself "operate a payment system" or hold customer funds as a payment service. Use KHQR/payment gateways to collect payment, then represent status changes in your order/payment tables. ¹⁷
- **Higher-regulatory surface model** (defer): if you intend to pool funds, run escrow, or offer stored value/wallet, you should expect deeper engagement with NBC licensing frameworks, operational controls, and audits. The MVP should avoid this unless licensing is already planned and funded. ¹⁶

Operational requirements for a Cambodia launch

Payments operations and reconciliation

KHQR-first, COD-first is the most Cambodia-appropriate operational baseline because KHQR is explicitly positioned as a standardised “single QR” enabling interoperability across banks/e-wallets via the Bakong payment network. ¹⁸

Operationally, KHQR integration requires: generating QR content (via KHQR SDK or EMVCo string generation), displaying it with expiration, polling/checking payment status via API, and updating merchant/buyer states. The NBC’s integration guide describes a product flow where the merchant sets an interval to check transaction status (with guidance that it should complete within seconds, otherwise fall back to manual receipt verification). ¹⁹

For COD, the commerce platform should treat the courier as the cash collector and your platform as the **state machine + reconciliation engine**. The minimum viable operational set is:

- clear COD eligibility rules by category and geography
- automatic COD confirmation messages to buyer/seller
- post-delivery cash reconciliation workflow (seller confirms receipt of COD remittance; admin can resolve disputes)

Fulfilment operations and SLAs

Even without owning logistics, you need **platform SLAs**. TikTok Shop’s fulfilment documentation demonstrates how SLAs are used to prevent late dispatch and enforce cancellation and seller accountability, including a process from payment confirmation to “awaiting shipment”, seller confirmation/packing, and dispatch tracking. ⁵

Your MVP should implement a simpler SLA policy and enforcement:

- maximum time to seller-confirm an order (separate for prepaid vs COD)
- maximum time to hand over to carrier
- minimum tracking update frequency (carrier scans or manual milestones)
- auto-cancellation rules for non-confirmation/non-dispatch (with buyer notification)

Customer support, disputes, and “after-sale” handling

The Cambodian e-commerce framework expects that consumers can see cancellation/withdrawal and return/refund terms as part of minimum information in electronic commerce. ¹¹

For MVP operations, a “structured dispute” model is required (not just chat):

- refund request reasons enumerated (wrong item/not delivered/defective/etc.)
- evidence attachments (photos, tracking screenshots)
- defined time windows (delivery date is the anchor)
- admin escalation and audit logs (who decided what and why)

KYC and seller verification operations

Two realities drive operational requirements: (1) you will be blamed by users for scams even if legally structured as a platform, and (2) intermediaries have specific duties when aware of unlawful activity.

6

Seller onboarding should therefore include:

- identity + business registration capture
- bank account details for payout
- restricted product checks (category whitelist for early phases)
- tiered limits for new sellers (order volume caps; longer payout holds)

Cambodia has a government data-exchange platform, CamDX ²⁰, positioned as a digital identity verification enabler; it is also used as part of e-commerce licensing system access via CamDigiKey-based login flows. ²¹

MVP architecture, explicit order state machine, and payout flows

Architecture overview and component diagram

A Codex-driven build should aim for a **modular monolith** (clear bounded contexts, single deployable initially) plus a small set of background workers. The platform must enforce order-state transitions server-side and maintain an immutable ledger for money movement.

```
flowchart LR
  subgraph Clients
    B[Buyer Web/App]
    S[Seller Dashboard]
    A[Admin Console]
  end

  subgraph Edge
    CDN[CDN + WAF]
    API[API Gateway / App Server]
  end

  subgraph Core
    AUTH[Auth + Roles]
    KYC[Seller Onboarding + KYC]
    CAT[Catalogue + Media]
    ORD[Orders + State Machine]
    PAY[Payments Adapter Layer]
    SHIP[Shipping Adapter Layer]
    LED[Ledger + Payouts]
    MOD[Moderation + Policy]
    NOTIF[Notifications]
  end

  subgraph Data
```

```

    DB[(PostgreSQL)]
    REDIS[(Redis/Queue)]
    OBJ[(Object Storage)]
    LOG[(Audit Logs)]
end

B --> CDN --> API
S --> CDN --> API
A --> CDN --> API

API --> AUTH
API --> KYC
API --> CAT
API --> ORD
API --> PAY
API --> SHIP
API --> LED
API --> MOD
API --> NOTIF

AUTH --> DB
KYC --> DB
CAT --> DB
CAT --> OBJ
ORD --> DB
PAY --> DB
SHIP --> DB
LED --> DB
MOD --> DB
NOTIF --> REDIS
API --> LOG

```

Compliance-driven rationale: keeping payments behind an adapter layer lets you prove that you are orchestrating payment status rather than operating a payment system, while still enabling NBC-standard KHQR flows and bank gateway integrations. ³

Data model: minimal tables you should not skip

The Law on Electronic Commerce includes record/evidence concepts and minimum consumer information duties, which make structured records non-optional. ²²

Minimum viable schema domains:

- **Identity & roles:** users, sessions, roles, permissions
- **Seller compliance:** sellers, seller_documents, seller_bank_accounts, verification_status, risk_tier
- **Catalogue/content:** products, product_variants (SKUs), inventory, price books, product_media, category, attributes
- **Orders:** orders, order_items, order_events (append-only), carts (optional after MVP), addresses
- **Payments:** payments, payment_attempts, payment_provider_events (raw webhook payload hashes), refunds
- **Shipping:** shipments, shipment_events, tracking mapping, courier accounts (if needed)

- **Ledger/payouts:** ledger_accounts, ledger_entries, payout_batches, payout_items, reserves/holds
- **Trust & moderation:** moderation_flags, enforcement_actions, disputes, returns
- **Audit:** admin_actions, immutable audit log

Explicit order state machine

Your order engine should use an enum plus a transition table enforced server-side. The state model below is designed to support both COD and prepaid KHQR/gateway.

OrderState (single canonical state)

- **CREATED** (order exists; payment method selected)
- **PAYMENT_PENDING** (prepaid only; awaiting confirmation)
- **PAYMENT_CONFIRMED** (prepaid success; COD skips to seller confirmation)
- **SELLER_CONFIRMATION_PENDING**
- **SELLER_CONFIRMED**
- **PACKED**
- **HANDED_TO_CARRIER**
- **IN_TRANSIT**
- **DELIVERED**
- **COMPLETED** (after dispute window closes)

Terminal/cancellation/after-sale states

- **CANCELLED_BY_BUYER** (only before seller confirmation or before shipment, per policy)
- **CANCELLED_BY_SELLER**
- **AUTO_CANCELLED_TIMEOUT** (payment or confirmation timeout)
- **RETURN_REQUESTED**
- **RETURN_APPROVED** / **RETURN_REJECTED**
- **RETURN_IN_TRANSIT** / **RETURN_RECEIVED**
- **REFUND_PENDING** / **REFUND_SUCCEEDED** / **REFUND_FAILED**
- **DISPUTE_OPEN** / **DISPUTE_RESOLVED**

Transition rules (examples)

- **CREATED** → **PAYMENT_PENDING** (prepaid)
- **PAYMENT_PENDING** → **PAYMENT_CONFIRMED** (provider webhook + reconciliation)
- **CREATED** → **SELLER_CONFIRMATION_PENDING** (COD)
- **PAYMENT_CONFIRMED** → **SELLER_CONFIRMATION_PENDING**
- **SELLER_CONFIRMATION_PENDING** → **SELLER_CONFIRMED** (seller accepts)
- **SELLER_CONFIRMED** → **PACKED** → **HANDED_TO_CARRIER** → **IN_TRANSIT** → **DELIVERED**
- **DELIVERED** → **COMPLETED** after a defined “cooldown/dispute window” (policy-driven)

This mirrors the operational emphasis seen in TikTok Shop’s fulfilment flow (payment confirmation drives “awaiting shipment”; seller then confirms/packs/ships with SLA implications). ⁵

Payout flow design with risk holds and reserves

TikTok Shop uses settlement periods tied to delivery date (T) and can apply different settlement periods (e.g., 3, 8, 15 days in some markets) and can hold a reserve for 30 days based on risk/returns. ²³ You can adopt the concept while tuning the numbers to your Cambodia realities.

Key constraint from Cambodian law: do not inadvertently become an unlicensed “electronic payment” operator. The law frames licensing obligations around operating payment systems/providing payment services/issuing payment instruments. ¹⁶ Therefore, the MVP payout design should be ledger-based and provider-integrated rather than wallet-based.

Recommended MVP payout model

- Maintain an internal ledger per seller and per platform fee account.
- Credit seller ledger when an order becomes **economically complete** (e.g., delivered + no dispute or delivered + hold elapsed).
- Payouts are executed via bank transfer/manual processing initially, moving to provider payout APIs later.

Two operational payout paths

1) **Prepaid (KHQR / gateway)** - Customer pays via KHQR or gateway. - Payment provider signals success (webhook/pushback); platform sets `PAYMENT_CONFIRMED`. PayWay's docs describe a flow where PayWay sends a pushback notification to the merchant endpoint after payment completion. ²⁴ - After delivery, start `HOLD_WINDOW` (risk-tier dependent). - On hold expiry: create ledger entries and include in next payout batch.

2) **COD** - Order is accepted with COD method. - Courier collects cash and remits to seller (or to the platform only if you have a controlled COD contract—defer for MVP). - Platform uses delivery confirmation (carrier scan or seller confirmation) to begin the hold window for platform fee settlement and chargebacks/disputes. - Seller payout is effectively “already paid” by cash; platform revenue collection occurs via seller billing, future payout offsets, or separate settlement.

Why holds are required even in lite MVP

Because consumer-facing return/refund/cancellation terms must be disclosed, and because your platform's fraud exposure concentrates around delivery disputes and counterfeit/undelivered claims. The e-commerce law also addresses platform/intermediary responsibilities around illegal content and data protection, making traceability and controlled settlement operationally important. ²⁵

Roadmap, timelines, and success metrics

Timelines below assume a small to medium build team using Codex with one product owner and one ops lead; **team size is unspecified**, so treat durations as ranges.

flowchart TD

```
P0[Foundation: compliance + architecture<br/>2-3 weeks] --> P1[Phase one (V1): marketplace core<br/>8-12 weeks]
P1 --> P2[Phase two (V2): scale + trust + automation<br/>8-10 weeks]
P2 --> P3[Phase three (V3): live selling + affiliates + optimisation<br/>>12-16 weeks]
```

Phase one scope and targets

Scope (V1) must deliver: seller onboarding (with document capture), catalogue, buyer feed + PDP, checkout (COD + KHQR), order-state machine, shipment/tracking fields, basic admin moderation, ledger with holds, and basic dispute/refund handling.

Minimum success metrics (definitions + initial targets)

- **Seller activation:** $\geq 30\%$ of approved sellers publish ≥ 10 SKUs within 7 days of approval.
- **Order completion rate:** $\geq 90\%$ of placed orders reach `DELIVERED` (excluding buyer-cancelled).
- **On-time dispatch:** $\geq 80\%$ of seller-confirmed orders reach `HANDED_TO_CARRIER` within your V1 SLA window (you set the window, then enforce it).
- **Payment confirmation latency (prepaid):** $p95 < 2$ minutes from buyer “paid” to `PAYMENT_CONFIRMED` (webhook + reconciliation). PayWay’s design explicitly relies on merchant-side confirmation via pushback and/or reconciliation API calls. ²⁶
- **Dispute rate:** $< 2\%$ of delivered orders enter `DISPUTE_OPEN` during the initial window.

Phase two scope and targets

Scope (V2): automate reconciliation, improve fraud controls, add vouchers/promotions, add structured returns centre, add courier integrations beyond manual entry, add seller performance scoring, and tighten content moderation.

Targets:

- reduce cancellation and dispute rates by 20–30% relative to V1 baseline (baseline must be measured),
- improve on-time dispatch to $\geq 90\%$,
- improve payout automation coverage where legally safe (e.g., provider payout APIs if available).

Phase three scope and targets

Scope (V3): livestream commerce, creator/affiliate commission system, stronger recommendation ranking, deeper logistics automation, and, only if planned, a controlled wallet/escrow model subject to NBC licensing analysis.

Targets:

- measurable uplift in conversion attributable to live/affiliate surfaces,
- stable payout loss rates (refund/chargeback) within defined risk budget,
- operational SLA parity across top provinces/cities.

Integration options and prioritised vendor links

Payments: KHQR, bank gateways, and card rails

KHQR via Bakong (priority for “pay by scan”)

KHQR is described by NBC materials as a standardised QR payment specification intended to promote interoperability; it “only requires a single QR” to receive payment from mobile apps and is implemented via the Bakong payment network. ²⁷ The NBC integration guide describes building dynamic QR flows where you generate KHQR via SDK or EMVCo generation and check transaction status via API. ¹⁹

ABA PayWay (priority for online checkout + webhook confirmation)

ABA Bank ²⁸’s PayWay developer suite documents purchase APIs, reconciliation endpoints, refunds, and dynamic QR/payment link options, and its sandbox PDF describes the merchant flow and pushback notifications. ²⁹

ACLEDA gateways (priority alternative for cards + KHQR + deeplinks)

ACLEDA Bank ³⁰ describes an e-commerce payment gateway for online selling that supports multiple acceptance methods (including KHQR and international cards) and outlines merchant qualification and

registration requirements. ³¹ ACLEDA's Toanchet Pay portal exposes sandbox functions including "KHQR String", "KHQR Web View", "Deeplink" and "Get Transaction Status", signalling multiple integration styles. ³²

Wing (priority for merchant network reach; integration details to be confirmed)

Official Wing merchant materials describe accepting payments from banks using Wing KHQR and provide merchant-facing tooling; however, public technical integration documentation is not clearly accessible in the sources retrieved here (site fetch constraints), so treat Wing as a commercial integration that may require a direct partner process. ³³

(Design implication: implement a payment-adapter interface so Wing can be added when partner APIs/specs are secured.)

Logistics: J&T, on-demand couriers, and national post

Nationwide parcel courier

Entity["company","J&T Express","courier | cambodia"] operates in Cambodia and maintains an API platform (documentation is presented under J&T API, with a commercial onboarding step requiring contacting an agent and agreements). ³⁴ This fits a V2 integration once volumes justify a formal contract.

Urban on-demand delivery

Grab ³⁵ offers GrabExpress for business with a portal-oriented delivery management model, suitable for same-day deliveries in major cities. ³⁶

National post option

Cambodia Post ³⁷ provides tracking and business solutions; its published materials include prohibited/dangerous item limitations that you can mirror in your category restrictions and shipping policy enforcement. ³⁸

Local logistics networks

Vireak Buntham Express ³⁹ is a known logistics operator with partner-facing pages, but detailed public API documentation is not evident in the retrieved sources; treat as a partner integration or manual workflow option initially. ⁴⁰

KYC / verification tools

Cambodia's licensing ecosystem uses CamDigiKey login flows; the broader CamDX platform has been positioned in official communications as supporting digital identity verification use cases. ⁴¹ For MVP, plan for "documents-first" KYC with the option to integrate CamDX-based verification where commercially and technically accessible. ⁴²

Prioritised vendor shortlist table

Priority	Domain	Vendor	Best for MVP	Integration style evidence	Key risks
High	KHQR	NBC Bakong KHQR	Pay-by-scan; interoperable QR	SDK/EMVCo generation + status check flow documented	Requires careful reconciliation + expiry handling ⁴³

Priority	Domain	Vendor	Best for MVP	Integration style evidence	Key risks
High	Gateway	ABA PayWay	Web checkout + webhook/ pushback + refunds	Purchase API + pushback webhook in docs	Domain/IP whitelisting; careful signature/ hash implementation ⁴⁴
Medium	Gateway	ACLEDA E-Commerce / Toanchet	Alternative gateway + KHQR/deeplink patterns	Merchant qualifications; sandbox functions list	Commercial onboarding; multi-rail complexity ⁴⁵
Medium	Wallet/ QR	Wing	Merchant presence + KHQR flows	Merchant materials mention Wing KHQR	Public API specs not retrieved; partner process likely ³³
High	Logistics	J&T Express	Nationwide shipping, potential COD logistics	API platform + onboarding steps documented	Contracting/ mapping effort; Cambodia-specific endpoints may differ ³⁴
Medium	Logistics	GrabExpress	City delivery, fast fulfilment	Business portal documented	Coverage and item constraints vary by market ³⁶
Medium	Logistics	Cambodia Post	Broad coverage, business solutions	Official site + restrictions published	Tracking/ integration may require third-party aggregation for APIs ⁴⁶
Medium	KYC	CamDX	Identity verification pathway	Platform positioned for identity verification; used in e-services access patterns	Access/ commercial model may require governmental coordination ²¹

Vendor links (for convenience)

KHQR / Bakong (NBC)
<https://bakong.nbc.gov.kh/en/>
<https://bakong.nbc.gov.kh/download/KHQR/integration/QR%20Payment%20Integration.pdf>
<https://bakong.nbc.gov.kh/download/KHQR/integration/KHQR%20SDK%20Document.pdf>

 ABA PayWay
<https://developer.payway.com.kh/>

<https://checkout-sandbox.payway.com.kh/plugins/payway-v2-sandbox.pdf>

ACLEDA

https://www.acledabank.com.kh/kh/eng/ps_ebecommerce

<https://toanchetpay.acledabank.com.kh/toanchetpay/sandbox>

Wing

<https://www.wingbank.com.kh/en/business/payments-collections/wing-merchant>

Logistics

<https://developer.jet.co.id/>

<https://www.grab.com/kh/en/business/express/>

<https://cambodiapost.com.kh/home>

E-commerce licensing / legal framework

<https://ecommercelicensing.moc.gov.kh/>

<https://commerce-cambodia.com/wp-content/uploads/2021/06/eCommerceLawEN.pdf>

<https://digitaleconomy.gov.kh/ecommerce?lang=en>

Developer-ready Codex prompt set with inputs, outputs, acceptance criteria, and snippets

The prompt set below is designed so each Codex run produces a **reviewable diff** with tests and acceptance checks. It assumes you are building a single repo with three frontends (buyer, seller, admin) and one backend.

Prompt pack table

Build step	Codex prompt (concise)	Expected inputs	Expected outputs	Acceptance criteria
Schema	"Create Prisma schema for marketplace core (users, sellers, products/SKUs, orders, payments, shipments, ledger, moderation, audit). Use integer minor units for money. Enums for statuses. Add indexes + constraints."	List of required tables; supported currencies	<code>schema.prisma</code> + migration + seed plan	<code>prisma migrate</code> succeeds; referential integrity enforced; money fields are integers
Auth	"Implement OTP/phone + role-based auth (buyer/seller/admin). Add session management, rate limits, and audit events on sensitive actions."	SMS provider choice (unspecified) → stub	Auth endpoints + middleware + tests	Login works; rate limits enforced; admin routes protected

Build step	Codex prompt (concise)	Expected inputs	Expected outputs	Acceptance criteria
Seller onboarding	"Implement seller onboarding workflow with document upload, verification status, risk tier, and admin approval queue."	Document types list	Seller UI + admin approval UI + DB changes	Seller cannot list without approval; all actions audited
Catalogue	"Implement products + variants + inventory; listing validation for mandatory consumer disclosures (price, currency, return policy, seller contact channel)."	Category list (initial)	CRUD APIs + buyer feed endpoints	Invalid listings rejected; feed loads under target latency
Checkout	"Implement cart/checkout supporting COD and prepaid. Create order + lock inventory atomically. Generate payment intent for prepaid."	Payment providers chosen	API + UI + integration stubs	No oversell under concurrency; order state correct
Order state machine	"Implement strict order-state enum + transition guard. Create append-only <code>order_events</code> . Reject illegal transitions server-side."	Transition matrix	Guard library + tests	100% tests for transition matrix; events recorded
Admin/moderation	"Build admin console: seller approvals, product moderation, dispute tooling, enforcement actions, audit logs."	Policy rules	Admin pages + API endpoints	All admin actions logged; moderation changes tracked
Payments	"Implement payment adapter interface with providers: KHQR (dynamic QR generation + status check) and PayWay (purchase + webhook). Store raw provider events hashed."	Provider credentials	Adapter modules + webhook endpoints	Webhooks verified; idempotent handling; reconciliation job works

Build step	Codex prompt (concise)	Expected inputs	Expected outputs	Acceptance criteria
Shipping	"Implement shipping adapter interface. V1: manual tracking entry. V2 hooks for courier APIs. Track shipment events."	Courier list	Shipment endpoints + seller UI	Seller can move to shipped with tracking; admin can override
Payouts	"Implement ledger accounting and payout batches with hold windows by risk tier. Support COD 'already collected' path. Provide payout reports."	Hold policy parameters	Ledger entries + payout services + reports	Balances reconcile; payout batch export works; negative balance handled
Seed data	"Create seed script for dev: admin user, demo sellers, sample products. Include Khmer/English examples."	Seed quantities	<code>seed.ts</code> / fixtures	<code>pnpm prisma db seed</code> produces usable demo
Tests	"Add integration tests for checkout, payment webhook idempotency, order transitions, and ledger posting."	Test stack choice	Test suite + CI config	Tests pass in CI; coverage thresholds met
Deployment	"Add docker-compose (Postgres, Redis). Add production Dockerfile, env templates, migration runbooks, and basic observability hooks."	Hosting target (unspecified)	Compose + docs + scripts	One-command local spin-up; health checks; rollback plan documented

Sample snippets requested

Prisma enum + core fields (illustrative)

```
enum OrderState {
  CREATED
  PAYMENT_PENDING
  PAYMENT_CONFIRMED
  SELLER_CONFIRMATION_PENDING
  SELLER_CONFIRMED
  PACKED
  HANDED_TO_CARRIER
  IN_TRANSIT
}
```

```

DELIVERED
COMPLETED
CANCELLED_BY_BUYER
CANCELLED_BY_SELLER
AUTO_CANCELLED_TIMEOUT
RETURN_REQUESTED
RETURN_APPROVED
RETURN_REJECTED
RETURN_IN_TRANSIT
RETURN_RECEIVED
REFUND_PENDING
REFUND_SUCCEEDED
REFUND_FAILED
DISPUTE_OPEN
DISPUTE_RESOLVED
}

model Order {
  id          String    @id @default(cuid())
  buyerId    String
  sellerId    String
  state       OrderState @default(CREATED)

  currency    String    // "KHR" or "USD" (validate at app layer)
  subtotalMinor Int
  shippingMinor Int
  discountMinor Int
  totalMinor  Int

  paymentMethod String // "COD" | "KHQR" | "PAYWAY" (enum in app)
  createdAt     DateTime @default(now())
  updatedAt     DateTime @updatedAt

  events       OrderEvent[]
  items        OrderItem[]
}

model OrderEvent {
  id          String    @id @default(cuid())
  orderId     String
  fromState   OrderState
  toState     OrderState
  actorType   String    // "BUYER" | "SELLER" | "ADMIN" | "SYSTEM"
  actorId     String?
  metadata    Json?
  createdAt   DateTime @default(now())

  order       Order @relation(fields: [orderId], references: [id])
  @@index([orderId, createdAt])
}

```


API route skeleton (idempotent webhook handler pattern)

```
// apps/api/src/routes/webhooks/payway.ts
import type { Request, Response } from "express";

export async function paywayWebhook(req: Request, res: Response) {
  // 1) Parse payload
  const payload = req.body;

  // 2) Compute idempotency key (provider transaction id + status + amount)
  const idemKey = `${payload.tran_id}:${payload.status}:${payload.apv ?? ""}`;

  // 3) If already processed => 200 OK
  // 4) Verify signature / shared secret if available (provider-specific)
  // 5) Persist raw event hash + minimal fields
  // 6) Transition order state with guard:
  //     PAYMENT_PENDING -> PAYMENT_CONFIRMED (success)
  // 7) Enqueue reconciliation job
  res.status(200).json({ ok: true });
}
```

Order-state guard (explicit transition table)

```
// packages/core/src/orderStateMachine.ts
export type OrderState =
  | "CREATED"
  | "PAYMENT_PENDING"
  | "PAYMENT_CONFIRMED"
  | "SELLER_CONFIRMATION_PENDING"
  | "SELLER_CONFIRMED"
  | "PACKED"
  | "HANDLED_TO_CARRIER"
  | "IN_TRANSIT"
  | "DELIVERED"
  | "COMPLETED"
  | "CANCELLED_BY_BUYER"
  | "CANCELLED_BY_SELLER"
  | "AUTO_CANCELLED_TIMEOUT"
  | "RETURN_REQUESTED"
  | "RETURN_APPROVED"
  | "RETURN_REJECTED"
  | "RETURN_IN_TRANSIT"
  | "RETURN_RECEIVED"
  | "REFUND_PENDING"
  | "REFUND_SUCCEEDED"
  | "REFUND_FAILED"
  | "DISPUTE_OPEN"
  | "DISPUTE_RESOLVED";
```

```

const ALLOWED: Record<OrderState, Set<OrderState>> = {
  CREATED: new Set(["PAYMENT_PENDING", "SELLER_CONFIRMATION_PENDING",
"CANCELLED_BY_BUYER"]),
  PAYMENT_PENDING: new Set(["PAYMENT_CONFIRMED", "AUTO_CANCELLED_TIMEOUT"]),
  PAYMENT_CONFIRMED: new Set(["SELLER_CONFIRMATION_PENDING",
"CANCELLED_BY_SELLER"]),
  SELLER_CONFIRMATION_PENDING: new Set(["SELLER_CONFIRMED",
"CANCELLED_BY_SELLER"]),
  SELLER_CONFIRMED: new Set(["PACKED"]),
  PACKED: new Set(["HANDED_TO_CARRIER"]),
  HANDED_TO_CARRIER: new Set(["IN_TRANSIT"]),
  IN_TRANSIT: new Set(["DELIVERED"]),
  DELIVERED: new Set(["COMPLETED", "RETURN_REQUESTED", "DISPUTE_OPEN"]),
  COMPLETED: new Set([]),

  CANCELLED_BY_BUYER: new Set([]),
  CANCELLED_BY_SELLER: new Set([]),
  AUTO_CANCELLED_TIMEOUT: new Set([]),

  RETURN_REQUESTED: new Set(["RETURN_APPROVED", "RETURN_REJECTED"]),
  RETURN_APPROVED: new Set(["RETURN_IN_TRANSIT"]),
  RETURN_REJECTED: new Set(["DELIVERED", "DISPUTE_OPEN"]),
  RETURN_IN_TRANSIT: new Set(["RETURN_RECEIVED"]),
  RETURN_RECEIVED: new Set(["REFUND_PENDING"]),
  REFUND_PENDING: new Set(["REFUND_SUCCEEDED", "REFUND_FAILED"]),
  REFUND_SUCCEEDED: new Set([]),
  REFUND_FAILED: new Set(["REFUND_PENDING", "DISPUTE_OPEN"]),

  DISPUTE_OPEN: new Set(["DISPUTE_RESOLVED"]),
  DISPUTE_RESOLVED: new Set(["COMPLETED"]),
};

export function assertTransition(from: OrderState, to: OrderState) {
  if (!ALLOWED[from].has(to)) throw new Error(`Illegal transition: ${from} -
> ${to}`);
}

```

Compliance-sensitive build notes for Codex implementation

- Seller-facing checkout/return/refund policies must be represented as structured fields because Cambodian e-commerce law requires disclosure of payment methods and cancellation/ withdrawal/return/exchange/refund details as minimum information in electronic commerce.
11
- Your “payments adapter” must be capable of supporting KHQR generation and status checking via NBC-described flows (SDK/EMVCo + status check) and gateway webhooks (PayWay pushback), while keeping your platform as the commerce orchestrator rather than the payment operator.
47

- Admin audit logs are not optional because the legal framework places duties on intermediaries once aware of illegal content/activity and because consumer protection enforcement can require proof of actions taken. ⁴⁸

¹ ⁵ ³⁰ https://seller-us.tiktok.com/university/essay?knowledge_id=6837879804970754

https://seller-us.tiktok.com/university/essay?knowledge_id=6837879804970754

² ³ ⁶ ⁷ ⁸ ⁹ ¹¹ ¹³ ¹⁴ ¹⁶ ²² ²⁵ ²⁸ ⁴⁸ <https://commerce-cambodia.com/wp-content/uploads/2021/06/eCommerceLawEN.pdf>

<https://commerce-cambodia.com/wp-content/uploads/2021/06/eCommerceLawEN.pdf>

⁴ ¹² https://digitaleconomy.gov.kh/images/fintech/pdf_figure_1.2/Law-on-Consumer-Protection.pdf

https://digitaleconomy.gov.kh/images/fintech/pdf_figure_1.2/Law-on-Consumer-Protection.pdf

¹⁰ <https://www.soksiphana.com/wp-content/uploads/2021/06/Notification-on-E-commerce-Permit-or-License-ENSSA20210607.pdf>

<https://www.soksiphana.com/wp-content/uploads/2021/06/Notification-on-E-commerce-Permit-or-License-ENSSA20210607.pdf>

¹⁵ ⁴¹ **E-commerce Licensing**

https://ec-user.moc.gov.kh/?utm_source=chatgpt.com

¹⁷ ¹⁹ ²⁰ ³⁹ ⁴³ ⁴⁷ <https://bakong.nbc.gov.kh/download/KHQR/integration/QR%20Payment%20Integration.pdf>

<https://bakong.nbc.gov.kh/download/KHQR/integration/QR%20Payment%20Integration.pdf>

¹⁸ ²⁷ ³⁵ <https://bakong.nbc.gov.kh/download/KHQR/integration/KHQR%20Content%20Guideline%20v.1.3.pdf>

<https://bakong.nbc.gov.kh/download/KHQR/integration/KHQR%20Content%20Guideline%20v.1.3.pdf>

²¹ ³⁷ **The NBC and the “Techo” Startup Center of the Ministry ...**

https://www.nbc.gov.kh/english/news_and_events/news_info.php?id=652&utm_source=chatgpt.com

²³ https://seller-th.tiktok.com/university/essay?knowledge_id=6837814452946689&lang=en

https://seller-th.tiktok.com/university/essay?knowledge_id=6837814452946689&lang=en

²⁴ ²⁶ ⁴⁴ <https://checkout-sandbox.payway.com.kh/plugins/payway-v2-sandbox.pdf>

<https://checkout-sandbox.payway.com.kh/plugins/payway-v2-sandbox.pdf>

²⁹ <https://developer.payway.com.kh/>

<https://developer.payway.com.kh/>

³¹ ⁴⁵ https://www.acledabank.com.kh/kh/eng/ps_ebecommerce

https://www.acledabank.com.kh/kh/eng/ps_ebecommerce

³² <https://toanchetpay.acledabank.com.kh/toanchetpay/sandbox>

<https://toanchetpay.acledabank.com.kh/toanchetpay/sandbox>

³³ **Wing Merchant App | Wingbank Cambodia**

https://www.wingbank.com.kh/en/business/payments-collections/wing-merchant?utm_source=chatgpt.com

³⁴ **J&T API**

<https://developer.jet.co.id/>

³⁶ **Grab for Business – Express | Grab KH**

<https://www.grab.com/kh/en/business/express/>

³⁸ ⁴⁶ **Home | Cambodia-Post**

<https://cambodiapost.com.kh/>

40 **Vireak Buntham Express**

https://vireakbuntham.com/partner?utm_source=chatgpt.com

42 **CamDX**

https://digitaleconomy.gov.kh/our-support?lang=en&utm_source=chatgpt.com